sys.all_columns (Transact-SQL)

Applies to: SQL Server · Azure SQL Database · Azure SQL Managed Instance · Azure Synapse Analytics · Analytics Platform System (PDW) · SQL analytics endpoint in Microsoft Fabric · Warehouse in Microsoft Fabric · SQL database in Microsoft Fabric

Shows the union of all columns belonging to user-defined objects and system objects.

Expand table

Column name	Data type	Description
`object_id`	**int**	ID of the object to which this column belongs.
`name`	**sysname**	Name of the column. Is unique within the object.
`column_id`	**int**	ID of the column. Is unique within the object. Column IDs might not be sequential.
`system_type_id`	**tinyint**	ID of the system-type of the column.
`user_type_id`	**int**	ID of the type of the column as defined by the user. To return the name of the type, join to the sys.types catalog view on this column.
`max_length`	**smallint**	Maximum length (in bytes) of the column. `-1` = Column data type is **varchar(max)**, **nvarchar(max)**, **varbinary(max)**, or **xml**. For **text** columns, the `max_length` value is `16` or the value set by `sp_tableoption 'text in row'`.
`precision`	**tinyint**	Precision of the column if numeric-based; otherwise, `0`.
`scale`	**tinyint**	Scale of the column if numeric-based; otherwise, `0`.
`collation_name`	**sysname**	Name of the collation of the column if character-based; otherwise, `NULL`.
`is_nullable`	**bit**	`1` = Column is nullable.
`is_ansi_padded`	**bit**	`1` = Column uses `ANSI_PADDING ON` behavior if character, binary, or variant.

Column name	Data type	Description
		`0` = Column isn't character, binary, or variant.
`is_rowguidcol`	bit	`1` = Column is a declared `ROWGUIDCOL`.
`is_identity`	bit	`1` = Column has identity values
`is_computed`	bit	`1` = Column is a computed column.
`is_filestream`	bit	`1` = Column is declared to use FILESTREAM storage.
`is_replicated`	bit	`1` = Column is replicated.
`is_non_sql_subscribed`	bit	`1` = Column has a non-SQL Server subscriber.
`is_merge_published`	bit	`1` = Column is merge-published.
`is_dts_replicated`	bit	`1` = Column is replicated by using SSIS.
`is_xml_document`	bit	`1` = Content is a complete XML document. `0` = Content is a document fragment, or the column data type isn't XML.
`xml_collection_id`	int	Non-zero if the column's data type is **xml** and the XML is typed. The value is the ID of the collection containing the column's validating XML schema namespace. `0` = no XML schema collection.
`default_object_id`	int	ID of the default object, regardless of whether it's a stand-alone sys.sp_bindefault, or an in-line, column-level `DEFAULT` constraint. The `parent_object_id` column of an inline column-level default object is a reference back to the table itself. `0` = No default.
`rule_object_id`	int	ID of the stand-alone rule bound to the column by using `sys.sp_bindrule`. `0` = No stand-alone rule. For column-level `CHECK` constraints, see sys.check_constraints.
`is_sparse`	bit	`1` = Column is a sparse column. For more information, see Use sparse columns.

Column name	Data type	Description
`is_column_set`	**bit**	`1` = Column is a column set. For more information, see Use column sets.

`generated_always_type`	**tinyint**	Identifies when the column value is generated (always `0` for columns in system tables):
		
		`0` = `NOT_APPLICABLE`
		`1` = `AS_ROW_START`
		`2` = `AS_ROW_END`
		`7` = `AS_TRANSACTION_ID_START`
		`8` = `AS_TRANSACTION_ID_END`
		`9` = `AS_SEQUENCE_NUMBER_START`
		`10` = `AS_SEQUENCE_NUMBER_END`
		
		For more information, see Temporal tables.
		
		Applies to: SQL Server 2016 (13.x) and later versions, and SQL Database. `7`, `8`, `9`, `10` only applies to SQL Database.

`generated_always_type_desc`	**nvarchar(60)**	**Applies to**: SQL Server 2016 (13.x) and later versions, and SQL Database.
		
		Textual description of the `generated_always_type` value (always `NOT_APPLICABLE` for columns in system tables)
		
		`NOT_APPLICABLE`
		`AS_ROW_START`
		`AS_ROW_END`
		
		Applies to: SQL Server 2022 (16.x) and later versions, and SQL Database
		
		`AS_TRANSACTION_ID_START`
		`AS_TRANSACTION_ID_END`
		`AS_SEQUENCE_NUMBER_START`
		`AS_SEQUENCE_NUMBER_END`

`ledger_view_column_type`	**tinyint**	If not `NULL`, indicates the type of a column in a ledger view:
		
		`1` = `TRANSACTION_ID`
		`2` = `SEQUENCE_NUMBER`
		`3` = `OPERATION_TYPE`
		`4` = `OPERATION_TYPE_DESC`
		
		For more information on database ledger, see Ledger.

Column name	Data type	Description
		Applies to: SQL Server 2022 (16.x) and later versions, and SQL Database.

`ledger_view_column_type_desc`	**nvarchar(60)**	If not `NULL`, contains a textual description of the type of a column in a ledger view:
		
		`TRANSACTION_ID`
		`SEQUENCE_NUMBER`
		`OPERATION_TYPE`
		`OPERATION_TYPE_DESC`
		
		Applies to: SQL Server 2022 (16.x) and later versions, and SQL Database.

`is_dropped_ledger_column`	**bit**	Indicates a ledger table column that was dropped. **Applies to**: SQL Server 2022 (16.x) and later versions, and SQL Database
`vector_dimensions`	**int**	Indicates how many dimensions the vector has. **Applies to**: SQL Server 2025 (17.x) and later versions, and SQL Database
`vector_base_type`	**tinyint**	Indicates the data type used to store vector dimensions values. `0` = 32-bit (single-precision) float `1` = 16-bit (half-precision) float [1] **Applies to**: SQL Server 2025 (17.x) and later versions, and SQL Database
`vector_base_type_desc`	**nvarchar(10)**	Contains the textual description of the data type used to store vector dimensions values. **Applies to**: SQL Server 2025 (17.x) and later versions, and SQL Database

[1] For more information, see Half-precision floating-point format.

Permissions

The visibility of the metadata in catalog views is limited to securables that a user either owns, or on which the user was granted some permission. For more information, see Metadata visibility configuration.

Related content

- Object catalog views (Transact-SQL)
- System catalog views (Transact-SQL)
- Querying the SQL Server System Catalog FAQ
- sys.columns (Transact-SQL)
- sys.system_columns (Transact-SQL)
- sys.computed_columns (Transact-SQL)

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